

1. Executive Summary & Overview

MediCare Family is a full-stack web application designed to solve medication mismanagement in multi-generational households. It centralizes prescription schedules, prevents conflicting and duplicate doses, enables instant 1-click dose tracking, and provides 7-day adherence analytics.

Key Objectives Solved:

- **Multi-Member Routines:** Individual profiles for elderly grandparents, parents, and children under one family login.
- **Zero Duplicates:** Strict backend logic prevents overlapping prescription names and duplicate daily timings.
- **Interactive Accountability:** 1-click dose logging (Taken / Skip / Undo) with real-time adherence scoring and streak tracking.

2. Complete Technology Stack

Layer	Technologies / Libraries	Description & Purpose
Frontend UI	React 18 + TypeScript + Vite	Modern reactive single-page application with type safety and fast compilation.
Styling & Design	Tailwind CSS	Custom health-oriented color tokens, accessible contrast, and responsive layout.
Visual Effects	Lucide Icons, Framer Motion, Confetti	Micro-animations, status icons, and celebration cues on completed doses.
Analytics & Charts	Recharts	Interactive Bar and Line Charts for scheduled vs. taken doses over 7 to 14 days.
Backend Server	Node.js + Express (TypeScript)	RESTful API architecture with route controllers, validation, and JWT auth middleware.
Database & ORM	Prisma ORM + SQLite	Type-safe database querying, declarative schema, and cascading deletion.
Security	JWT + Bcrypt.js	Encrypted user passwords (10 salt rounds) and stateless token validation.

3. Step-by-Step Page & Feature Operation Guide

A. Authentication & Demo Mode ([/login](#))

- **1-Click Instant Demo Login:** Click "Sign In as Demo Family (1-Click)" to instantly explore pre-configured data for **The Miller Family** (Arthur, Sarah, Leo) with 8 active medications and 7-day historical logs.
- **Standard Login:** Enter email ([demo@medicare.family](#)) and password ([demo1234](#)) and click "Sign In to Dashboard".
- **Register New Family:** Click "Don't have an account yet? Create one", input family name, email, and password to initialize a fresh, isolated workspace.

B. Dashboard & Daily Schedule ([/](#))

- **Daily Progress Overview:** Real-time progress bar showing today's taken vs. pending doses and adherence percentage.
- **Date Switcher:** Navigate rapidly using "Yesterday", "Today", "Tomorrow", or select any custom date via the calendar picker.
- **Multi-Filter:** Filter visible doses by specific family member or by time of day (Morning: 00:00-11:59, Afternoon: 12:00-16:59, Evening: 17:00-23:59).
- **1-Click Dose Actions:**
 - **"Mark Taken":** Marks dose completed, records exact timestamp, and triggers celebration confetti.
 - **"Skip":** Logs dose as intentionally skipped.
 - **"Undo":** Reverts accidental clicks back to pending status.

C. Family Members Management ([/members](#))

- **Add Member:** Click "+ Add Member", enter name, relation (Grandfather, Mother, Child, etc.), age, and choose a custom avatar theme.
- **Individual Member Profile ([/members/:id](#)):** View personalized adherence rates, all active prescriptions, and quickly add medicines specifically for that member.
- **Edit / Delete:** Update member details or remove members with automatic cascade cleanup of their medications and logs.

D. Medicine & Prescription Scheduling ([/medicines](#))

- **Add Prescription:** Click "+ Add Medicine", choose member, enter medication name, dosage (e.g., [500mg](#) , [1 tablet](#)), instructions, and start/end dates.
- **Multi-Timing Presets:** Add one or multiple daily dosing times using fast presets (*Morning 08:00, Noon 13:00, Night 20:30*) or custom 24-hr entries.
- **Backend Duplicate Validation:** Automatically blocks identical medicine names for the same member during overlapping date intervals.

E. Adherence Analytics ([/adherence](#))

- **7-Day & 14-Day View:** Toggle historical windows to monitor consistency trends.
- **Streak Counter:** Shows consecutive days of 100% medication compliance (e.g. 🔥 *6-Day Perfect Streak*).
- **Interactive Charts:** Visual comparison of daily scheduled doses vs. taken doses across all members.
- **Weekly Status Grid:** Per-member matrix breakdown indicating daily completed vs. missed doses.

4. Quick Action Cheat Sheet

Goal / Action	Location in Application
View Today's Medicine List	Dashboard (/)
Mark Medicine Taken	Click " Mark Taken " (Green button) on any dose card
Add New Family Profile	Top Navbar → Click "+ Add Member "
Schedule New Medication	Top Navbar → Click "+ Add Medicine "
Inspect Charts & Streaks	Sidebar → Click " Adherence "
Switch User / Sign Out	Top Navbar → Click " Logout "